2. Let $f : A \to B$ be a function defined by $f(x) = \frac{x}{2} - 1$, where $A = \{2,4,6,10,12\}$, $B = \{0,1,2,4,5,9\}$. Represent f by

   (i) set of ordered pairs (ii) a table (iii) an arrow diagram (iv) a graph

3. Represent the function $f = \{(1,2),(2,2),(3,2),(4,3),(5,4)\}$ through

   (i) an arrow diagram (ii) a table form (iii) a graph

4. Show that the function $f : \mathbb{N} \to \mathbb{N}$ defined by $f(x) = 2x - 1$ is one-one but not onto.

5. Show that the function $f : \mathbb{N} \to \mathbb{N}$ defined by $f(m) = m^2 + m + 3$ is one-one function.

6. Let $A = \{1,2,3,4\}$ and $B = \mathbb{N}$. Let $f : A \to B$ be defined by $f(x) = x^3$ then,

   (i) find the range of f (ii) identify the type of function

7. In each of the following cases state whether the function is bijective or not. Justify your answer.

   (i) $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 2x + 1$ (ii) $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 3 - 4x^2$

8. Let $A = \{-1,1\}$ and $B = \{0,2\}$. If the function $f : A \to B$ defined by $f(x) = ax + b$ is an onto function? Find a and b.

9. If the function f is defined by $f(x) = \begin{cases} x + 2; & x > 1 \\ 2; & -1 \le x \le 1 \\ x - 1; & -3 < x < -1 \end{cases}$ find the values of

   (i) $f(3)$ (ii) $f(0)$ (iii) $f(-1 \cdot 5)$ (iv) $f(2) + f(-2)$

10. A function $f : [-5,9] \to \mathbb{R}$ is defined as follows:

$$f(x) = \begin{cases} 6x + 1; & -5 \le x < 2 \\ 5x^2 - 1; & 2 \le x < 6 \\ 3x - 4; & 6 \le x \le 9 \end{cases}$$

    Find (i) $f(-3) + f(2)$ (ii) $f(7) - f(1)$ (iii) $2f(4) + f(8)$ (iv) $\frac{2f(-2) - f(6)}{f(4) + f(-2)}$

11. The distance S an object travels under the influence of gravity in time t seconds is given by $S(t) = \frac{1}{2}gt^2 + at + b$ where, (g is the acceleration due to gravity), $a > 0$, b are constants. Verify wheather the function $S(t)$ is one-one or not.

12. The function 't' which maps temperature in Celsius (C) into temperature in Fahrenheit (F) is defined by $t(C) = F$ where $F = \frac{9}{5}C + 32$. Find,

    (i) $t(0)$ (ii) $t(28)$ (iii) $t(-10)$

    (iv) the value of C when $t(C) = 212$

    (v) the temperature when the Celsius value is equal to the Farenheit value.

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